

## Quantitative Research Design

### 1. Identify and define the following:

The purpose of this study is to investigate whether there are statistically significant differences in test scores obtained from students who have been exposed to three different types of educational environments (traditional educational instruction, instruction enhanced by additional work with computers and the Internet, and instruction enhanced by additional training in multi-cultural sensitive) in their classes over the course of an academic year.

### Research Design

A quantitative experimental design will be employed, and participants are randomly assigned into treatment and control groups. Traditional educational instruction is the control group. Instruction enhanced by additional work with computers and the Internet and instruction enhanced by additional training in multi-cultural sensitive are the experimental groups (Gay, Mills & Airasian, 2006, p. 254).

### *Variables*

The independent variable is a variable denoting different treatment conditions, with different groups being treated differently. In this study, there are three independent group variables and they are defined by the “type of educational environment”: a) 1 = traditional; b) 2 = computer and the Internet; and c) 3 = multi-cultural sensitive.

The dependent variable is a variable which denotes a measurement on a coherent, scaled attribute, as defined by a score on that attribute. In this study, the dependent variable is defined as the “test score”.

### *Sample*

Approximately 60 participants (i.e. students) will be randomly obtained from each school. Participants will include only those whose parents or guardians grant permission for use of student records. In this study, approximately 180 participants will be needed to carry out the research.

2. Assuming that the dependent variable is measured on an interval scale, what statistical procedure would be most appropriate to evaluate the research question?

The statistical procedure for comparing three independent groups (i.e. type of educational environment) on the dependent variable (i.e. test score) is called one-way ANOVA in SPSS. This is simply comparing the means of groups in order to make inferences about the population. One-way ANOVA is also called single factor analysis of variance, because there is only one independent variable or factor. The independent variable has nominal levels (Gay et al., 2006, p. 359; Morgan, Leech, Gloeckner, & Barrett, 2007, p. 157). In this study, we will use the one-way ANOVA procedure, because we have one independent variable with three independent groups.

The level of measurement of the independent variable is defined as nominal, because there are three unordered categories (i.e. type of educational environment), and numerical values are simply to name the attribute uniquely.

The level of measurement of the dependent variable is measured on an interval scale where distances between levels are equal, but no true zero.

3. If the correct statistical test used to evaluate these data yields a p value (or the Sig. value in SPSS) of .040, what does that result indicate about the groups?

Either 5 out of 100 or .05 is the typical alpha level that researchers use to access whether the null hypothesis should be rejected or not (Gay et al., 2006, p. 343; Morgan et al., 2007, p. 92).

For  $p = .04$ , the result of this study is not likely due to chance alone, because  $p$  value is less than .05 (i.e.  $p < .05$ ). The result suggests that there is a significant difference between the three types of education environments (or treatment conditions) on test scores. However, we do not know which specific groups of means are significantly different, unless we do follow-up tests. Multiple comparison can be used to determine which means are significantly different from which other means (Morgan et al., 2007, p. 160).

4. If the correct statistical test used to evaluate these data yields a  $p$  value (or the Sig. value in SPSS) of .895, what does that result indicate about the groups?

For  $p = .895$ , the result of this study is likely due to chance alone, because  $p$  value is larger than .05 (i.e.  $p > .05$ ). The result suggests that there is not a significant difference in average student test scores among the three types of educational environments (or treatment conditions) and that there is no evidence to suggest the difference and no further test is needed.

## Part B: Qualitative Research Design

### Research Design

#### *Purpose*

The purpose of this study is to investigate similarities and differences in perceptions of performance of students who have, in their classes, been exposed to three different types of educational environments over the course of an academic year.

*Research Question*

Are there similarities and differences between the types of environments (traditional educational instruction, instruction enhanced by additional work with computers and the Internet, and instruction enhanced by additional training in multi-cultural sensitive) in regard to the perceptions of performance of students?

*Method*

A qualitative experimental design will be employed. Phenomenological methodology will be used to investigate similarities and differences between the types of environments, in regard to the perceptions of students' performance. This method reveals the meanings of our everyday existence. According to van Manen (1990), it is "the fulfillment of our human nature: to become more fully who we are" (p. 12). This perspective suggests that individuals reflect much of their sentiments regarding research inquiries and data analysis that observe the lives of each individual within their own unique contexts (Creswell, 2007, p. 69).

In this study, the investigator/researcher (etic reporting) collects descriptions of behavior through observations, interviews, documents, and artifacts (Creswell, 2007, p. 131).

*Site and Participant Selection*

In this study, we will stratify a sample of 10 students, in their classes, from each of the three types of educational environments (traditional educational instruction, instruction enhanced by additional work with computers and the Internet, and instruction enhanced by additional training in multi-cultural sensitive). The larger sample size may be used to ensure that smaller (or harder to reach) subgroups are still likely to be represented. However, larger sample size may waste time, resources, and money. According to Creswell (2007), he stated that in

phenomenology, he saw the number of participants range from 1 to 325 (Creswell, 2007, p. 126). In this study, emic reporting is the participant themselves.

A gate keeper may be use to get in contact with the potential participants for the study. Creswell (2007) suggested “bracketing” prior to the interview process.

#### *Consent Form*

In order to ensure the protection of human subjects, all participants must have their parents or guardians review and sign a consent form prior to their participation in the study.

#### *Interview Questions*

In this study, we will ask three open-ended questions to all participants from the three types of educational environments. The following questions are addressed:

1. How would you describe your school’s environment?
2. What did you already know about multicultural diversity?
3. What kinds of learning experiences do you have in your classroom such as art, music, math, science, social studies, computer, and the Internet?

#### Analyze data

The phenomenological data analysis tends to follow specific structured methods. Creswell (2007) suggests three general approaches to analysis—Madison, Huberman and Miles, and Wolcott. In this study, we will use data analysis strategies by Wolcott, which suggests sketching ideas, reducing codes to themes, relating categories to analytic framework in literature, and displaying the data (Creswell, 2007, p. 149).

In addition, researchers can use several techniques discussed by Patten (2005) to establish the dependability and trustworthiness of the data. Specific techniques include: a) data triangulation; b) methods triangulation; c) researcher triangulation; d) team of researchers with

diverse backgrounds; e) interobserver agreement; f) peer review; g) auditor; and h) member checking.

In this study, we will use the process of inductive analysis, and it basically means that categories and patterns emerge from the data rather than being imposed on data prior to data collection (McMillan, 2001, p. 462). We will also use open coding and axial coding. In addition, we will use triangulation to establish the dependability and trustworthiness of the data.

Before we can begin the analysis of the data, we will need to transcribe the interviews into text data. Initially, we will use hand coding of the transcripts with paper and pencil methods, and a codebook is to construct, revise, and use throughout the data analysis to record emerging concepts and subcategories.

Qualitative data includes structured text, unstructured text, audio recordings, video recordings, etc. Creswell (2007) suggests that Computer-Assisted Qualitative Data Analysis Software (CAQDAS) is very helpful in computerizing the process of analyzing text and imaging data (Creswell, 2007, p. 168). Creswell goes on to list the following advantages of using computers as a tool to accomplish qualitative research:

1. Provides an organized storage system.
2. Can help researcher locate material quickly and easily.
3. Encourages researchers to look closely at the data line by line.
4. Concept mapping feature enables the researcher to visualize relationships by drawing a visual model.
5. Allows a researcher to easily retrieve memos associated with codes, themes, or documents (Creswell, 2007, p. 165).

2. Describe how each of the following key terms apply in the design of your study:

*Bracketing or Reflexivity*

Bracketing, reflectivity, or epoche are concepts in which researchers set aside their experiences, as much as possible, to take a fresh perspective toward the phenomenon under examination (Creswell, 2007, p. 59). These ideas suggest that researchers should set aside what they know about the student, so it does not influence their study results.

### *Open Coding*

Open coding is the first step in the data analysis process. It involves taking the interview transcriptions from the students and segmenting them into categories of information. This suggests that researchers should try to develop a small number of categories, and slowly reduce the number of categories, until that data becomes a theme for the study (Creswell, 2007, p. 64).

### *Axial Coding*

Axial coding is the second step, following open coding. In this step, researchers take the categories of open coding, and identify one as a central phenomenon about the types of environments, and then return to the database to identify the cause of the phenomenon (Creswell, 2007, p.237). This suggests that categories are systematically developed and linked with subcategories.

### *Triangulation*

This is the process of using multiple method data collection strategies and data sources to obtain a more complete picture of what is being studied, and to cross check information. The strength of qualitative research lies in collecting information in many ways, rather than relying solely on one. Often two or more methods can be used in such a way that the weakness of one is compensated by the strength of the other (Creswell, 2007, p. 208). For example, interviews with students may be used to contribute to our understanding of what we observe happening in a learning environment.

*Gate keeper*

A gate keeper is an informant or a participant, that allows the researcher to observe and spend time talking to the participants in the study (Creswell, 2007, p. 68). A gate keeper is essentially a person who helps the researcher to get in contact with the students.

*Emic reporting*

Emic (participant) reporting is an insider participant who observes the study and retells the story about what happened during the interview (Creswell, 2007, p. 72).

*Etic reporting*

Etic (researcher) reporting is very similar to Emic reporting. Etic reporting is an outsider participant, who explains the observer's role to the participants (i.e. students) (Creswell, 2007, p. 72).



## References

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